

BobaLink “ Development Guide

Revision: 1 **Last modified:** 2026-06-10T21:00:00Z **Scope:** Architecture, layout, testing, and contribution rules for BobaLink (extension/). **Authority:** extension/package.json, extension/wxt.config.ts, docs/browser_extension/IMPLEMENTATION_PLAN.md, docs/browser_extension/Status.md (Rev 4), project CLAUDE.md + constitution/.

Accuracy note (Â§11.4.6): the architecture and commands below reflect the real tree at HEAD 2011810. Phases still open are marked (**in progress**).

Stack

- **WXT** (wxt ^0.19.0) “ the WebExtension build framework; generates the MV3 manifest from wxt.config.ts and bundles entrypoints into .output/.
 - **TypeScript** (^5.7.0, strict) “ npm run compile is tsc --noEmit.
 - **Vitest** (^2.1.8) + @vitest/coverage-v8 “ unit/integration/etc. tests.
 - **Playwright** (@playwright/test ^1.49.0) “ end-to-end (real MV3 extension load).
 - **ESLint** (^9.17.0) + Prettier “ lint/format.
 - Runtime dependency: webextension-polyfill.
 - Node “¥ 20, ESM ("type": "module").
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Architecture

The pipeline, end to end:

page DOM

â†’ parser	(magnet / bencode / .torrent â†’ SHA-1 infohash)
â†’ scanner	(link-scanner + text-scanner, MutationObserver, site s
â†’ orchestrator	(cross-scanner dedup by infohash)
â†’ background	(service-worker message router; context menus; command
â†’ boba-client	(POST to the Boba merge service on :7187; token header
â†’ queue	(offline FIFO; persists across SW restart; auth-inject

Module map (under extension/src/):

- **parsers** “ magnet URI parsing, bencode decoding, .torrent â†’ SHA-1 infohash (SHA-1(bencode(info)) via WebCrypto). Dedup key is the lowercase infohash.
- **scanners** “ link-scanner / text-scanner + content/scanner; stable infohash-based IDs; Shadow-DOM/iframe aware; 500 ms MutationObserver debounce.
- **orchestrator** “ runs the scanners and deduplicates across them.
- background/index.ts “ the service-worker message router: dispatches context-menu clicks (boba-link-send / boba-link-send-all / boba-link-send-group), keyboard commands (send-to-boba / scan-page / open-dashboard), and chrome.alarms keep-alive + periodic health checks.
- api/boba-client.ts “ the Boba merge-service client (Phase 4, **in progress**): posts magnets/URLs to :7187, presents the optional decrypted Boba API token as an

Authorization/X-Boba-Token header, with retry/backoff and an AbortController timeout.

- `api/queue.ts` – the offline FIFO queue; enqueues on send failure and flushes when the backend recovers; durable across service-worker restart.
- `tabgroups/index.ts` – dedupe + batch dispatch across a Chrome tab group, invoked from the background `MENU_SEND_GROUP` handler.
- `shared/` – `constants.ts`, `crypto.ts` (AES-GCM-256 / PBKDF2), `errors.ts`, `events.ts`, `logger.ts`, `storage.ts`, `utils.ts`.
- `options/options.ts` + `popup/` – the 7-tab options page and the detected-list popup with Send / Send-All.

WXT entrypoints (thin wrappers)

WXT discovers entrypoints under `extension/src/entrypoints/`:

```
src/entrypoints/
  background.ts      â†’ background.js service worker
  content.ts         â†’ content-scripts/content.js
  popup/index.html   â†’ popup.html
  options/index.html â†’ options.html
```

These are **thin wrappers** over the logic modules listed above – the heavy logic lives in `src/{parsers,scanners,background,api,options,popup,shared,...}/`, so the existing unit tests import the logic modules directly and stay valid regardless of WXT's entrypoint plumbing (Status.md Rev 4, "WXT build wiring – COMPLETE").

`wxt.config.ts` sets `srcDir: "src"`, `outDir: ".output"`, the MV3 manifest (least-privilege permissions, localhost-only `host_permissions`, the CSP, the three keyboard commands), and `@wxt-dev/auto-icons` to rasterize icon sizes.

Running tests

From `extension/`:

```
npx vitest run          # the full unit/integration/security/perf/etc. suite
npm run test            # == vitest run --coverage
npm run test:watch      # vitest in watch mode
npm run test:live       # vitest run --config tests/live/vitest.live.config
npm run test:e2e        # playwright test (real MV3 extension load)
```

- The Vitest corpus lives under `extension/tests/{unit,perf,stress,chaos,integration,security,e2e}` and alongside source in `src/**`. As of Status.md Rev 4 a same-session `npx vitest run` reported **Tests 413 passed (413)** across 37 spec files.
- `test:live` targets a live backend; live integration uses a `require_backend(7187)` guard that **SKIPS with reason** when `:7187` is down (never fail-open, §11.4.3/§11.4.68).
- `test:e2e` loads the real built extension via Playwright (`launchPersistentContext + --load-extension`, real id from the SW target). The MV3-load e2e is **operator-gated SKIP** in a headless sandbox where extension load is unsupported.

- Anti-bluff is the gate: each test asserts a user-observable outcome and must fail against a no-op stub (IMPLEMENTATION_PLAN.md §0.2). Challenges under challenges/ extension/ and challenges/ security/ drive the real orchestrator/client/audit end-to-end and are mutation-verified.

Type + lint, run by ci-ext.sh and individually:

```
npm run compile      # tsc --noEmit
npm run lint         # eslint src/ tests/ --ext .ts
npm run format:check # prettier --check "src/**/*.ts"
```

Constitution constraints (read before contributing)

BobaLink inherits the Boba CLAUDE.md and the constitution/ submodule. The load-bearing rules for this subproject:

- **Anti-bluff (§11.4).** Every test/Challenge must prove the feature works for the end user — assert user-observable outcomes (DB rows, response bodies, DOM text), and each test must FAIL against a no-op stub. The reference suite's 10 known bluffs are remediated, not adopted.
- **TDD (§11.4.43/§11.4.115).** Write the failing RED test first (reproducing the real defect / driving the real feature), watch it fail, then write minimal code to GREEN. "Test added after the fix" is a bluff.
- **NO CI/CD — ever (Hard Stop §1).** No .github/workflows/, no pipelines, no git hooks. Validation is the manual extension/ci-ext.sh.
- **No force-push (§11.4.113); fetch-first (§11.4.71); SSH remotes only.**
- **§11.4.10 credentials.** Delegate-by-default; no embedded/decryptable secret by default; no fixed/empty-passphrase decrypt; no real credential in any test/fixture/doc.
- **100% multi-type coverage (§11.4.27/§11.4.85).** unit/integration/e2e/security/ load/ scaling/chaos/stress/performance/benchmark/UI/UX + Challenges + HelixQA.
- **Mandatory code review (§11.4.125/§11.4.134/§11.4.142)** of every change, iterate-until-GO, before it is "complete".

The single real backend is the **Boba merge service on :7187** — never qBittorrent directly, never port 8080 (the reference :8080 values were all retargeted to :7187; see src/shared/constants.ts).

Where the docs / Status / ledger live

All under docs/browser_extension/:

- Status.md (+ .html/.pdf) — the authoritative per-phase status (Rev 4 is the real state).
- Status_Summary.md (+ siblings) — the two-audience summary companion.
- IMPLEMENTATION_PLAN.md (+ siblings) — the 9-phase master plan with the hard constraints, backend additions (BE-1/2/3), and adopt/refactor/rewrite map.
- coverage_ledger.md (+ siblings) — the feature — test-type coverage ledger.
- USER_GUIDE.md, INSTALL.md, DEVELOPMENT.md (this file) — the Phase-9 user/ install/dev docs (+ .html/.pdf siblings per §11.4.65).
- Deeper provenance: docs/browser_extension/_analysis/ and _plan/.

The script companion for the gate is `docs/scripts/ci-ext.md` (§11.4.18). HelixQA banks: `submodules/helixqa/banks/boba-bobalink.yaml` (symlinked into `challenges/helixqa-banks/`).